

PUBLICLY AVAILABLE SPECIFICATION

**Connectors for electrical and electronic equipment – Product requirements –
Part 3-126: Rectangular connectors – Detail specification for 5 pole power
connectors for industrial environments with push-pull locking**



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**Connectors for electrical and electronic equipment – Product requirements –
Part 3-126: Rectangular connectors – Detail specification for 5 pole power
connectors for industrial environments with push-pull locking**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

CONNECTORS FOR ELECTRICAL AND ELECTRONIC EQUIPMENT – PRODUCT REQUIREMENTS –

Part 3-126: Rectangular connectors – Detail specification for 5 pole power connectors for industrial environments with push-pull locking

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The text of this PAS is based on the following document:

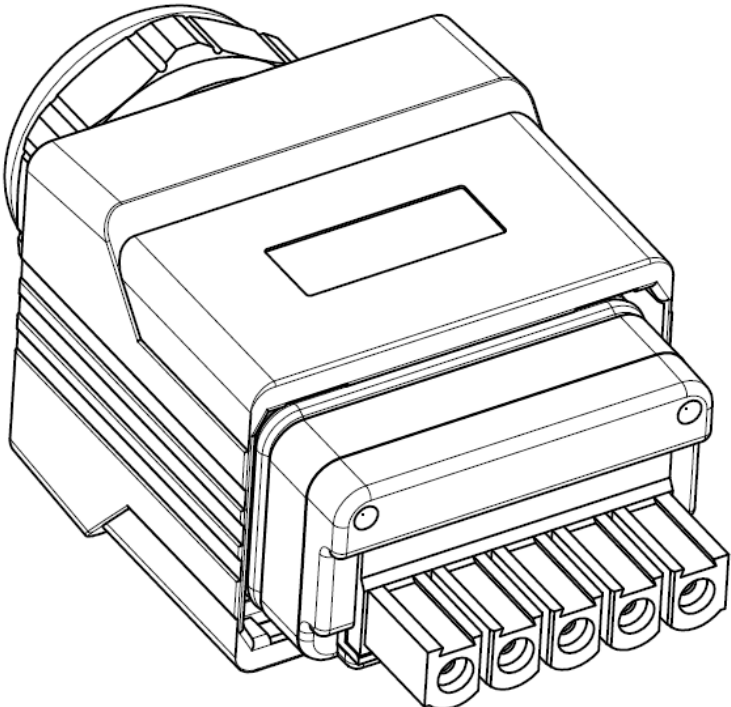
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INTRODUCTION

IEC SC 48B – Electrical connectors Specification available from: IEC General secretariat or from the addresses shown on the inside cover.	IEC 61076-3-126 Ed.1
ELECTRONIC COMPONENTS DETAIL SPECIFICATION in accordance with IEC 61076-1	
	Rectangular connectors Detail specification for power connectors for industrial environments with push-pull locking Male and female connectors Male and female contacts Rewirable – Non-rewirable
	Free cable connectors Straight and right angle connectors Fixed connectors Flange mounting Single hole mounting

CONNECTORS FOR ELECTRICAL AND ELECTRONIC EQUIPMENT – PRODUCT REQUIREMENTS –

Part 3-126: Rectangular connectors – Detail specification for 5 pole power connectors for industrial environments with push-pull locking

1 Scope

This document covers rectangular IP65/IP67 connectors with 5 poles for electric power supply up to 16 A. These connectors consist of fixed and free connectors, either rewirable or non-rewirable (for both portions). It uses the general function principles of the push-pull connector housing system described in IEC 61076-3-117 with IP65/IP67 degree of protection according to IEC 60529 for harsh applications.

Male connectors have 5 square 1 mm electric contacts, with 16 A rated current. Connectors according to this document are without breaking capacity COC according to IEC 61984, therefore they are not intended to be engaged or disengaged in normal use when live or under load, if not otherwise specified by the manufacturer.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-581:2008, *International Electrotechnical Vocabulary – Chapter 581: Electro-mechanical components for electronic equipment*

IEC 60068-1, *Environmental testing – Part 1: General and guidance*

IEC 60068-2-30, *Environmental testing – Part 2-30: Tests – Test Db: Damp heat, cyclic (12 h + 12 h cycle)*

IEC 60352-1, *Solderless connections – Part 1: Wrapped connections – General requirements, test methods and practical guidance*

IEC 60352-2, *Solderless connections – Part 2: Crimped connections – General requirements, test methods and practical guidance*

IEC 60352-3, *Solderless connections – Part 3: Solderless accessible insulation displacement connections – General requirements, test methods and practical guidance*

IEC 60352-4, *Solderless connections – Part 4: Solderless non-accessible insulation displacement connections – General requirements, test methods and practical guidance*

IEC 60352-5, *Solderless connections – Part 5: Press-in connections – General requirements, test methods and practical guidance*

IEC 60352-6, *Solderless connections – Part 6: Insulation piercing connections – General requirements, test methods and practical guidance*

IEC 60352-7, *Solderless connections – Part 7: Spring clamp connections – General requirements, test methods and practical guidance*

IEC 60352-8, *Solderless connections – Part 8: Compression mount connections – General requirements, test methods and practical guidance*

IEC 60512-1-1, *Connectors for electronic equipment – Tests and measurements – Part 1-1: General examination – Test 1a: Visual examination*

IEC 60512-1-2, *Connectors for electronic equipment – Tests and measurements – Part 1-2: General examination – Test 1b: Examination of dimension and mass*

IEC 60512-2-1, *Connectors for electronic equipment – Tests and measurements – Part 2-1: Electrical continuity and contact resistance tests – Test 2a: Contact resistance – Millivolt level method*

IEC 60512-2-2, *Connectors for electronic equipment – Tests and measurements – Part 2-2: Electrical continuity and contact resistance tests – Test 2b: Contact resistance – Specified test current method*

IEC 60512-2-5, *Connectors for electronic equipment – Tests and measurements – Part 2-5: Electrical continuity and contact resistance tests – Test 2e: Contact disturbance*

IEC 60512-2-6, *Connectors for electronic equipment – Tests and measurements – Part 2-6: Electrical continuity and contact resistance tests – Test 2f: Housing (shell) electrical continuity*

IEC 60512-3-1, *Connectors for electronic equipment – Tests and measurements – Part 3-1: Insulation tests – Test 3a: Insulation resistance*

IEC 60512-4-1, *Connectors for electronic equipment – Tests and measurements – Part 4-1: Voltage stress tests – Test 4a: Voltage proof*

IEC 60512-5-2, *Connectors for electronic equipment – Tests and measurements – Part 5-2: Current-carrying capacity tests – Test 5b: Current-temperature derating*

IEC 60512-6-1, *Connectors for electronic equipment – Tests and measurements – Part 6-1: Dynamic stress tests – Test 6a: Acceleration, steady state*

IEC 60512-6-2, *Connectors for electronic equipment – Tests and measurements – Part 6-2: Dynamic stress tests – Test 6b: Bump*

IEC 60512-6-3, *Connectors for electronic equipment – Tests and measurements – Part 6-3: Dynamic stress tests – Test 6c: Shock*

IEC 60512-6-4, *Connectors for electronic equipment – Tests and measurements – Part 6-4: Dynamic stress tests – Test 6d: Vibration (sinusoidal)*

IEC 60512-6-5, *Electromechanical components for electronic equipment – Basic testing procedures and measuring methods – Part 6: Dynamic stress tests – Section 5: Test 6e: Random vibration*

IEC 60512-8-1, *Connectors for electronic equipment – Tests and measurements – Part 8-1: Static load tests (fixed connectors) – Test 8a: Static load, transverse*

IEC 60512-8-2, *Connectors for electronic equipment – Tests and measurements – Part 8-2: Static load tests (fixed connectors) – Test 8b: Static load, axial*

IEC 60512-9-1, *Connectors for electronic equipment – Tests and measurements – Part 9-1: Endurance tests – Test 9a: Mechanical operation*

IEC 60512-9-2, *Connectors for electronic equipment – Tests and measurements – Part 9-2: Endurance tests – Test 9b: Electrical load and temperature*

IEC 60512-11-1, *Electromechanical components for electronic equipment – Basic testing procedures and measuring methods – Part 11: Climatic tests – Section 1: Test 11a – Climatic sequence*

IEC 60512-11-3, *Connectors for electronic equipment – Tests and measurements – Part 11-3: Climatic tests – Test 11c: Damp heat, steady state*

IEC 60512-11-4, *Connectors for electronic equipment – Tests and measurements – Part 11-4: Climatic tests – Test 11d: Rapid change of temperature*

IEC 60512-11-9, *Connectors for electronic equipment – Tests and measurements – Part 11-9: Climatic tests – Test 11i: Dry heat*

IEC 60512-11-10, *Connectors for electronic equipment – Tests and measurements – Part 11-10: Climatic tests – Test 11j: Cold*

IEC 60512-11-13, *Connectors for electronic equipment – Tests and measurements – Part 11-13: Climatic tests – Test 11n: Gas tightness, solderless wrapped connections*

IEC 60512-12-4, *Connectors for electronic equipment – Tests and measurements – Part 12-4: Soldering tests – Test 12d: Resistance to soldering heat, solder bath method*

IEC 60512-12-5, *Connectors for electronic equipment – Tests and measurements – Part 12-5: Soldering tests – Test 12e: Resistance to soldering heat, soldering iron method*

IEC 60512-13-1, *Connectors for electronic equipment – Tests and measurements – Part 13-1: Mechanical operation tests – Test 13a: Engaging and separating forces*

IEC 60512-13-2, *Connectors for electronic equipment – Tests and measurements – Part 13-2: Mechanical operation tests – Test 13b: Insertion and withdrawal forces*

IEC 60512-13-5, *Connectors for electronic equipment – Tests and measurements – Part 13-5: Mechanical operation tests – Test 13e: Polarizing and keying method*

IEC 60512-14-7, *Electromechanical components for electronic equipment – Basic testing procedures and measuring methods – Part 14: Sealing tests – Section 7: Test 14g: Impacting water*

IEC 60512-15-1, *Connectors for electronic equipment – Tests and measurements – Part 15-1: Connector tests (mechanical) – Test 15a: Contact retention in insert*

IEC 60512-16-5, *Connectors for electronic equipment – Tests and measurements – Part 16-5: Mechanical tests on contacts and terminations – Test 16e: Gauge retention force (resilient contacts)*

IEC 60512-15-6, *Connectors for electronic equipment – Tests and measurements – Part 15-6: Connector tests (mechanical) – Test 15f: Effectiveness of connector coupling devices*

IEC 60512-17-3, *Connectors for electronic equipment – Tests and measurements – Part 17-3: Cable clamping tests – Test 17c: Cable clamp resistance to cable pull (tensile)*

IEC 60512-17-4, *Connectors for electronic equipment – Tests and measurements – Part 17-4: Cable clamping tests – Test 17d: Cable clamp resistance to cable torsion*

IEC 60512-19-3, *Electromechanical components for electronic equipment – Basic testing procedures and measuring methods – Part 19: Chemical resistance tests – Section 3: Test 19c - Fluid resistance*

IEC 60529:1989, *Degrees of protection provided by enclosures (IP Code)*

IEC 60664-1:2007, *Insulation coordination for equipment within low-voltage systems – Part 1: Principles, requirements and tests*

IEC 60998-2-1:2002, *Connecting devices for low-voltage circuits for household and similar purposes – Part 2-1: Particular requirements for connecting devices as separate entities with screw-type clamping units*

IEC 60999-1:1999, *Connecting devices – Electrical copper conductors – Safety requirements for screw-type and screwless-type clamping units – Part 1: General requirements and particular requirements for clamping units for conductors from 0,2 mm² up to 35 mm² (included)*

IEC 60999-2, *Connecting devices – Electrical copper conductors – Safety requirements for screw-type and screwless-type clamping units – Part 2: Particular requirements for clamping units for conductors above 35 mm² up to 300 mm² (included)*

IEC 61076-1:2006, *Connectors for electronic equipment – Product requirements – Part 1: Generic specification*

IEC 61076-3:2008, *Connectors for electronic equipment – Product requirements – Part 3: Sectional specification for rectangular connectors*

IEC 61984:2008, *Connectors - Safety requirements and tests*

IEC 62430:2009, *Environmentally conscious design for electrical and electronic products*

IEC GUIDE 109, *Environmental aspects – Inclusion in electrotechnical product standards*

ISO 128-30:2001, *Technical drawings – General principles of presentation – Part 30: Basic conventions for views*

ISO 1302:2002, *Geometrical Product Specifications (GPS) – Indication of surface texture in technical product documentation*

ISO 11469:2016, *Plastics – Generic identification and marking of plastics products*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60050-581 as well as the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

IEC Electropedia: available at <http://www.electropedia.org/>

ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1 protective conductor PE

conductor provided for purposes of safety, for example protection against electric shock

Note 1 to entry: In an electrical installation, the conductor identified PE is normally also considered as protective earthing conductor.

[SOURCE: IEC 60050-581:2008, 581-27-26]

3.2 functional earth conductor FE

earthing conductor provided for functional earthing

Note 1 to entry: Functional earthing a point or points in a system or in an installation or in equipment, for purposes other than electrical safety.

[SOURCE: IEC 60050-195, Amendment 1: 2001, 195-01-13]

4 Technical information

4.1 Systems of levels

4.1.1 Performance levels

See Table 9.

4.1.2 Compatibility levels, according to IEC 61076-1:2006

Connectors according to this document are deemed to be intermateable.

4.2 Classification into climatic categories

Table 1 shows the climatic category.

Table 1 – Climatic category

Climatic category	Category temperature		Damp heat steady state		Days
	Lower °C	Upper °C	Temperature °C	Relative humidity %	
25/070/21	–25	+70	40	93	21

4.3 Clearance and creepage distances

Creepage and clearance distances shall meet the requirements given in 6.4.1.

4.4 Current-carrying capacity

See 6.4.3.

4.5 Marking

The marking of the connector and the package shall be in accordance with 2.7 of IEC 61076-1:2006.

5 Dimensional information

5.1 Common features

Connectors meant to comply with IP ratings per IEC 60529 shall be appropriately mated and sealed in order to meet the requirements detailed in the test schedules in 7.2.

5.2 Reference system

Coordination dimensions are dimensions without tolerances which indicate the boundary or centre-line references in order to allow arrangement.

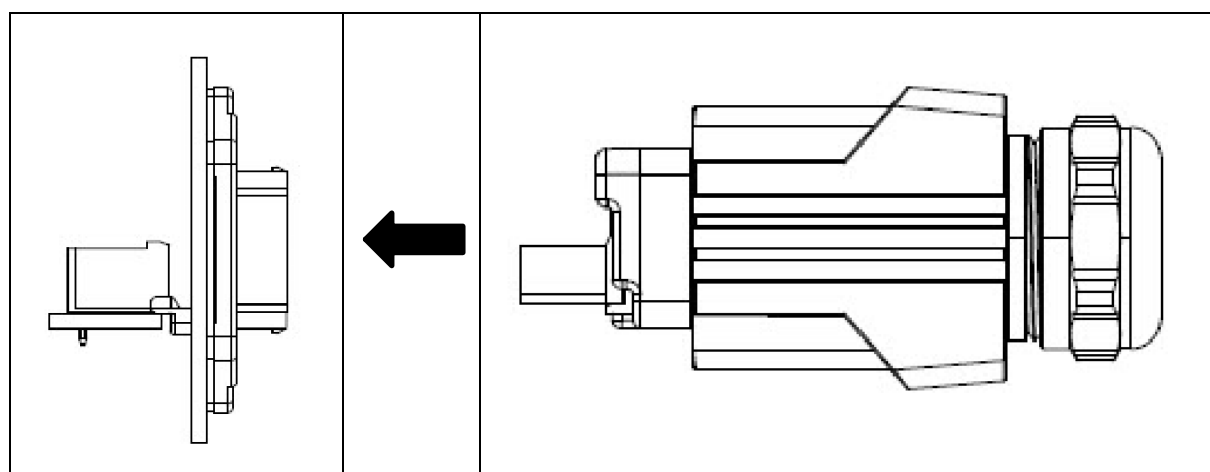
Dimensions are given in millimetres, drawings are shown in first angle projection according ISO 128-30. The shape of connectors may deviate from those shapes given in the following figures as long as the specified dimensions are not influenced.

5.3 Engagement (mating) information

5.3.1 Engaging (mating) direction

5.3.1.1 General

An arrow in Figure 1 indicates the mating direction.



IEC

Figure 1 – View showing mating direction

5.3.1.2 Contact levels and sequencing

For all interfaces, the contact sequence during the connection process shall be:

- Protective Earth (if any)
- Functional Earth (if any)
- Power contacts

During disconnection the order shall be reversed.

5.3.2 Perpendicular to the engaging (mating) direction

Not applicable.

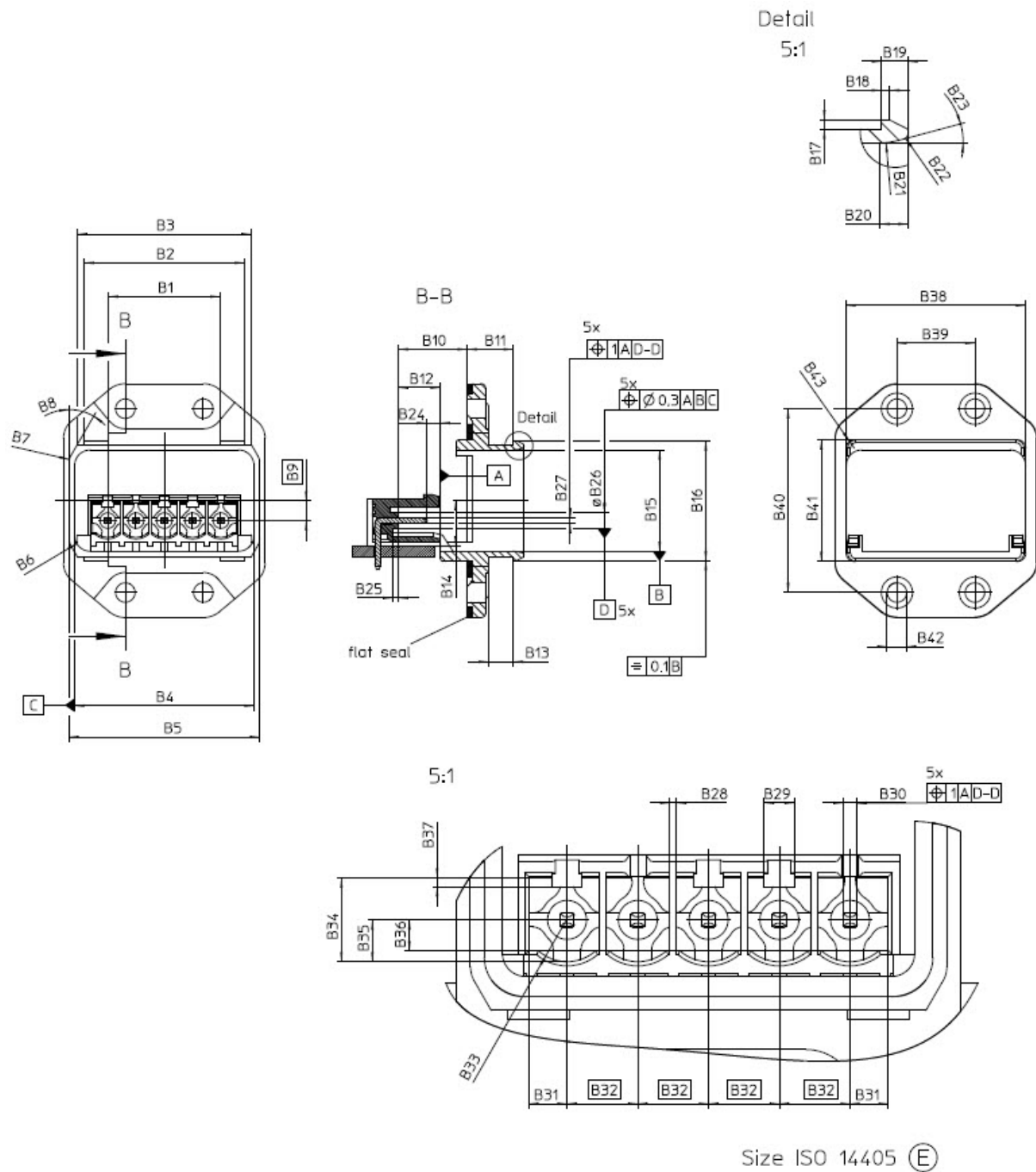
5.3.3 Inclination

Not applicable.

5.4 Fixed connectors

5.4.1 Dimensions

Figure 2 shows a fixed connector with male contacts and Table 2 shows the related dimensions.



IEC

Figure 2 – Fixed connector with male contacts

Table 2 – Dimensions of fixed connector

<i>Dimensions in millimetres</i>			
Letter	Minimum	Nominal	Maximum
B1	19,9	20	20,1
B2	28,7	28,8	28,9
B3	31,05	31,2	31,35
B4	32,15	32,2	32,25
B5	34,1	34,2	34,3
B6	R 2,75	R 2,8	R 2,85
B7	R 1,93	R 2	R 2,07
B8	29°	30°	31°
B9	-	3,6 1	-
B10	12,2	12,35	12,5
B11	8,25	8,3	8,35
B12	7,44	7,5	7,56
B13	4,4	4,5	4,6
B14	8,05	8,15	8,25
B15	18,15	18,2	18,25
B16	21,45	21,5	21,55
B17	0,6	0,65	0,8
B18	0,55	0,6	0,65
B19	1,85	1,9	1,95
B20	1,9	2	2,1
B21	R 4,9	R 5	R 5,1
B22	R 0,3	R 0,4	R 0,5
B23	14°	15°	16°
B24	1,4	2,4	2,6
B25	0,95	1	1,1
B26	2,73	2,8	2,87
B27	0,95	1	1,05
B28	0,40	0,5	0,55
B29	2,04	2,14	2,24
B30	0,95	1	1,05
B31	2,64	2,74	2,99
B32	-	5,08 1	-
B33	R 2,98	R 3,08	R 3,38
B34	5,95	6	6,2
B35	2,95	3	3,05
B36	2,1	2,2	2,35
B37	0,45	0,55	0,65
B38	32,1	32,25	32,4
B39	13,9	14	14,1
B40	32,9	33	33,1
B41	21,65	21,85	22,05
B42	3,4	3,5	3,6
B43	1,25	-	-
NOTE Theoretically precise dimension.			

5.4.2 Electrical terminations

The contact terminations shall be of any type.

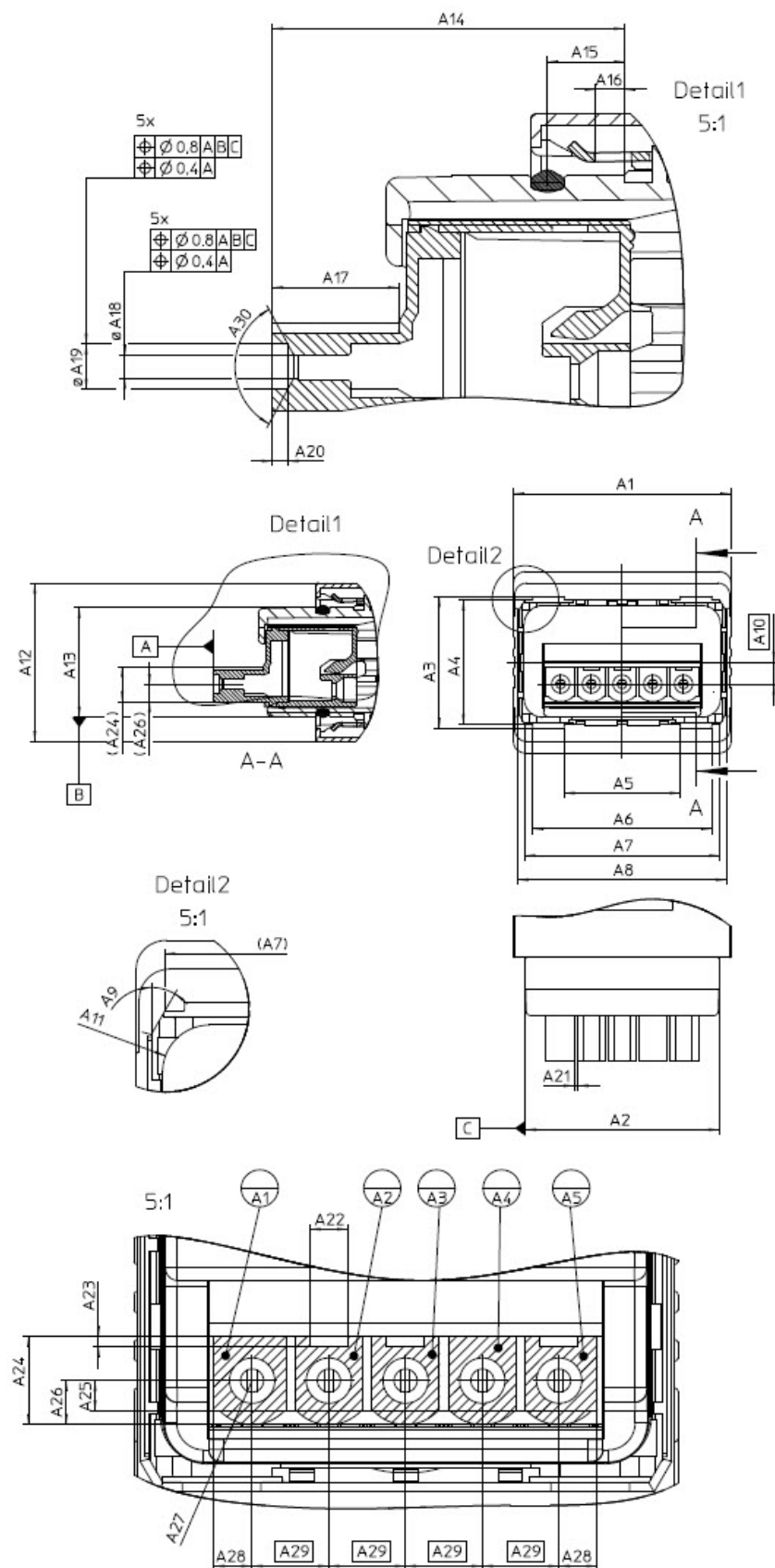
Solderless terminations shall be in accordance with the relevant part of the IEC 60352 series.

Screw-type or screwless-type terminations shall be in accordance with IEC 60999-1.

5.5 Free connectors

5.5.1 Dimensions

Figure 3 shows a free connector with female contacts and Table 3 shows the related dimensions.



Size ISO 14405 (E)

IEC

Figure 3 – Free connector with female contacts

Table 3 – Dimensions of free connector

<i>Dimensions in millimetres</i>			
Letter	Minimum	Nominal	Maximum
A1	35,95	36,2	36,45
A2	31,89	32,06	32,09
A3	21,7	-	-
A4	20,45	-	-
A5	-	-	19,35
A6	29,45	-	-
A7	31,45	-	-
A8	34,4	34,5	34,6
A9	29°	30°	31°
A10	-	3,6 ¹	-
A11	R 3,07	R 3,1	R 3,3
A12	-	-	26,8
A13	17,89	18,06	18,09
A14	23	23,2	23,4
A15	4,6	5,1	5,6
A16	2,15	2,25	2,45
A17	8,26	8,33	8,38
A18	1,48	1,53	1,58
A19	2,93	3	3,1
A20	0,95	1	1,1
A21	0,57	0,6	0,8
A22	2,5	2,6	2,7
A23	0,63	0,68	0,73
A24	5,73	5,83	5,88
A25	1,95	2	2,05
A26	2,89	2,92	2,95
A27	R 2,78	R 2,93	R 3,03
A28	2,1	2,52	2,57
A29	-	5,08 ¹	-
A30	120°	125	130°
NOTE Theoretically precise dimension.			

5.5.2 Electrical terminations

The contact terminations shall be of the following types: screw, crimp, piercing, insulation displacement, press-in or solder.

Solderless terminations shall be in accordance with the relevant part of the IEC 60352 series.

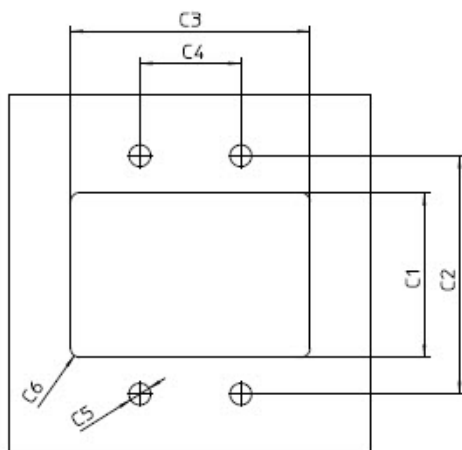
Screw-type or screwless-type terminations shall be in accordance with IEC 60999-1.

5.6 Accessories

Not applicable.

5.7 Mounting information for connectors

Figure 4 shows a panel cut out for the fixed connector and Table 4 shows the related dimensions.



IEC

Figure 4 – Mounting information – panel cut-out outline

Table 4 – Mounting dimensions

<i>Dimensions in millimetres</i>			
Letter	Minimum mm	Nominal mm	Maximum mm
C1	22,7	22,8	22,9
C2	32,9	33	33,1
C3	33,1	33,2	33,3
C4	13,9	14	14,1
C5	-	M 3	-
C6	-	-	1,25

5.8 Gauges

5.8.1 Sizing gauges and retention force gauges

Material: tool steel, hardened



= Surface roughness acc. to ISO 1302: $Ra_{\max} = 0,25 \mu\text{m}$, $Ra_{\min} = 0,15 \mu\text{m}$

Surface clean and free of grease.

Figure 5 shows gauge dimensions.

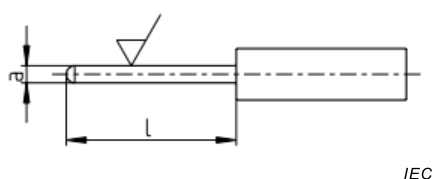


Figure 5 – Gauge dimensions

Table 5 shows gauge dimensions.

Table 5 – Gauge dimensions

Gauge	Mass g	Application	a mm	$I \text{ min}$ mm	Nom pin a mm
P51	-	Sizing	1,03 mm +0,01 / 0 mm	10	1,0
P52	25	Retention force	0,97 mm 0 / -0,01 mm	10	

5.8.2 Mechanical function, engaging/separating/insertion/withdrawal force gauges

Not applicable.

5.8.3 Probes

Not applicable.

5.8.4 Contact resistance gauge

Not applicable.

5.8.5 Test panel (for voltage proof test)

For all styles of fixed connectors, the panel cut-out shall be designed according to the drawings in 5.7 with cut-out dimensions 2 μm larger than the maximum permitted corresponding dimension of the fixed connector. The test panel shall have the maximum thickness.

For free connectors, the connectors shall be wrapped into a foil of aluminium. The foil shall be connected to PE.

5.8.6 Test panel (for EMC/crosstalk, etc.)

Not applicable.

6 Characteristics

6.1 General

For the ratings, see Table 6.

Table 6 – Ratings of connectors

Contacts	Rated voltage a.c. or d.c. V	Rated current A
5 (4+FE)	60	16

Insulation resistance: $10^8 \Omega \text{ min.}$

Climatic category: see Table 1

Contact spacing: see Clause 5

The Detail Quality Specification according to IEC 62197-2-1xx is not applicable.

6.2 Pin assignment and other definitions

See Clause 5.

6.3 Classification into climatic categories

Conditions: IEC 60068-1

Table 1 shows the climatic category.

The lowest and highest temperatures and the duration of the damp heat, steady state test should be selected from the preferred values stated in 2.2 of IEC 61076-1:2006.

6.4 Electrical characteristics

6.4.1 Creepage and clearance distances

Clearance and creepage distances shall be measured according to IEC 60512-1-2 with the following additional requirements.

For this connector clearance and creepage distances shall be measured only in mated position (connector without breaking capacity COC as defined in IEC 61984).

The connector has protection against electric shock by finger safety (IP2X or IPXXB), both in the mated or unmated condition, after mounting (see also 6.4.2.2 of IEC 61984:2008). For minimum values for clearance and creepage distances see Clause 5.

6.4.2 Voltage proof

The coordination of creepage and clearance distances has to be performed according IEC 60664-1.

The connector shall be tested according to the ratings in 6.1 and shall be mounted to the test panel specified in 5.8.5.

Conditions: IEC 60512-4-1, Test 4a

Standard atmospheric conditions

Mated connectors

For ratings see Table 6 and Table 8.

Application information: The permissible rated voltage depends on the application or specified safety requirements. Reductions in creepage or clearance distances may occur due to the printed board or wiring used, and shall duly be taken into account.

Table 7 shows rated impulse voltage and pollution degree.

Table 7 – Rated impulse voltage – Pollution degree

Rated impulse voltage kV	Pollution degree
1,5	3

Table 8 shows voltage proof.

Table 8 – Voltage proof

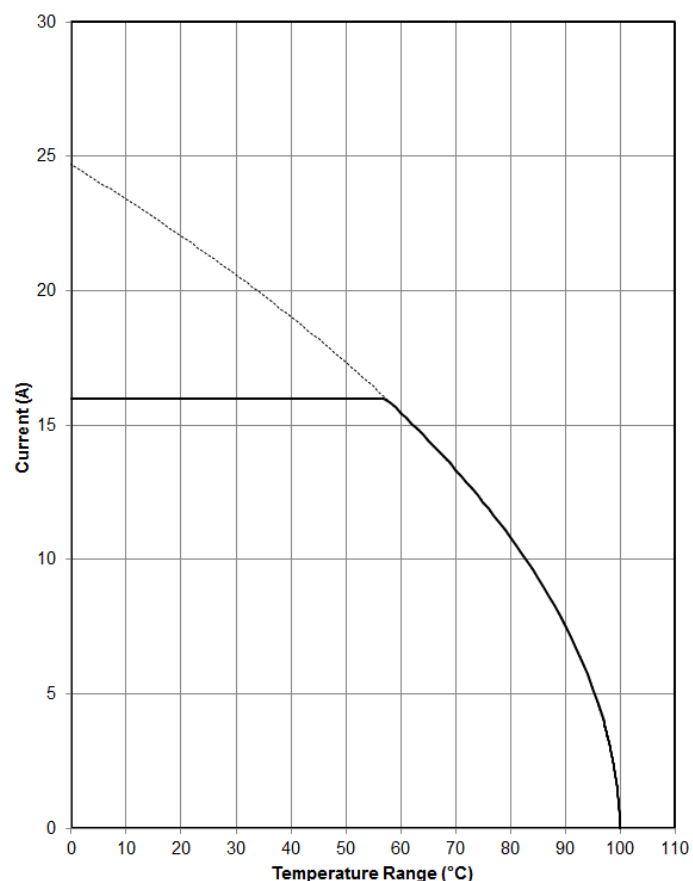
kV withstanding voltage			
Between contacts		Between contacts and metal-housing	
Fixed connectors	Free connectors	Fixed connectors	Free connectors
2,21	2,21	2,21	2,21

6.4.3 Current-carrying capacity

Conditions: IEC 60512-5-2, Test 5b

All contacts except FE and PE, connected in series.

The current-carrying capacity of connectors shall comply with the de-rating curve given in Figure 6 as a minimum requirement.



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NOTE 1 The maximum permissible current I for an ambient temperature between $T = -25\text{ °C}$ and $T = 40\text{ °C}$ is $I = 16\text{ A}$.

NOTE 2 The maximum permissible current I for an ambient temperature between $T = 40\text{ °C}$ and $T = 70\text{ °C}$ is $I = 16\text{ A} \cdot (1 - (T - 40)/35)^{0.5}$.

Figure 6 – Connector de-rating curve

6.4.4 Contact and shield resistance

Conditions: IEC 60512-2-1, Test 2a, IEC 60512-2-2, Test 2b and IEC 60512-2-6, Test 2f

Standard atmospheric conditions.

Conditions: IEC 60512-2-1, Test 2a

Mated connectors.

Measurement points: 40 mm behind the cable glands

Power contacts and FE contact: 10 mΩ max.

$U_{\max} \leq 20\text{ mV}$ (AC or DC)

$I_m \leq 100\text{ mA}$ (AC or DC)

Shield contact (if any): not applicable.

Among all conductors, difference between maximum and minimum:

10 mΩ max.

$R2_{\text{(after test)}} \leq R1_{\text{(before test)}} + 5\text{ mΩ}$

6.4.5 Insulation resistance

Conditions: IEC 60512-3-1, Test 3a, Method A

Standard atmospheric conditions

Mated connectors

Test voltage: 500 V d.c.

Each contact to all others: $10^8 \Omega \text{ min}$

6.4.6 Impedance

Not applicable.

6.4.7 Transmission characteristics

Not applicable.

6.5 Mechanical characteristics

6.5.1 Mechanical operation

Conditions: IEC 60512-9-1, Test 9a

Standard atmospheric conditions

Max. speed of operations = 10 mm/s

Rest: 30 s, unmated

Table 9 shows the number of mechanical operations.

Table 9 – Number of mechanical operations

Performance level	Mechanical operations
P21	100
other types	^a
^a Other mating cycles are upon agreement between manufacturer and user.	

6.5.2 Effectiveness of connector coupling devices

Conditions: IEC 60512-15-6, Test 15f

Standard atmospheric conditions

Applied force: 150 N

Test duration: 60s $\pm$ 5s

Force speed: $\leq 44,5 \text{ N/s}$

6.5.3 Engaging and separating forces (or insertion and withdrawal forces)

Conditions: IEC 60512-13-1 Test 13a

Coupling speed: $\leq 10 \text{ mm/s}$

All types, insertion and withdrawal: 60 N maximum

Initial torque test insertion: 1,0 Nm max. and withdrawal 0,7 Nm max.

After conditioning torque test insertion 2,0 Nm max. and withdrawal 1,2 Nm max.

6.5.4 Contact retention in insert

Not applicable.

6.5.5 Polarizing and coding method

See Table 10.

Conditions: IEC 60512-13-5, Test 13e

Standard atmospheric conditions

Table 10 – Total insertion force

Number of poles	Total insertion force N
5	Max. 90

6.6 Other characteristics

6.6.1 Vibration (method half-sine)

Conditions: IEC 60512-6-4, Test 6d

Frequency range: 10 Hz – 500 Hz

Amplitude: 0,35 mm

Frequency change: 1 octave/min.

Number of sweeps per axis: 10

Axes: 3 – x, y, z

Acceleration: 50 m/s²

Standard atmospheric conditions

Connectors in mated and locked position

The fixed and free connector shall be rigidly installed in a suitable fixture as specified in Figure 7.

Duration: 10 swaps each axis

Cable with maximum wire size or overmoulded cable.

Figure 7 shows a dynamic stress test arrangement.

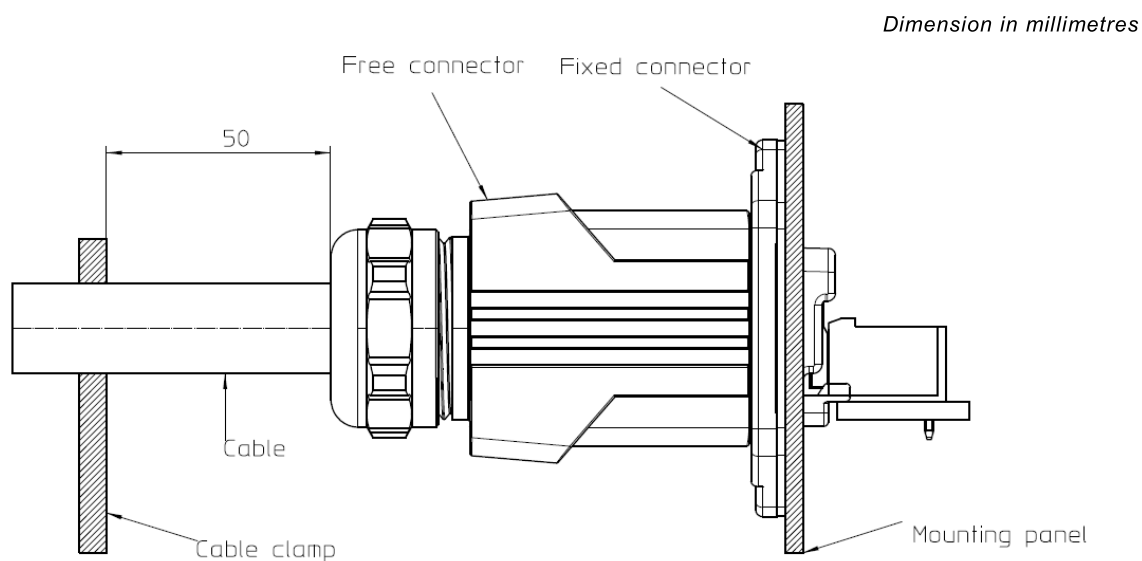


Figure 7 – Stress test arrangement

IEC

6.6.2 Shock (method half-sine)

Conditions: IEC 60512-6-3, Test 6c

Duration of the impact: 11 ms

Axes: 3 – x, y, z

Acceleration: 500 m/s²

Standard atmospheric conditions

Connectors in mated and locked position

The fixed and free connector shall be rigidly installed in a suitable fixture as specified in Figure 7.

Duration: 6 shocks per axis (3 positive axis, 3 negative axis)

Cable with maximum wire size or overmoulded cable.

6.6.3 Degree of protection provided by enclosures (IP-code)

Conditions: IEC 60529, test 14.2.5 and test 14.2.7 (second numeral) and

IEC 60529, IP6X test 6, Table 7 (first numeral)

Both IPX5 and IPX7 (second numeral) according to IEC 60529

Connectors in mated and locked position.

Fixed connector to be mounted according 5.8.5.

6.6.4 Screen and shielding properties

Not applicable.

6.6.5 Static load test

Conditions: IEC 60512-8-1, Test 8a.

A force of 50 N to be applied at the end of the free housing to load the coupling device with the maximum torque.

6.7 Environmental aspects

6.7.1 Marking of insulation material (plastics)

If applicable and possible, all plastic material should be marked according to ISO 11469 to ease recycling.

6.7.2 Design/use of material

The design has to take into account the relevant IEC guides for designing products (IEC 62430) and the use of material (IEC Guide 109) with regard to the environment.

7 Test schedule

7.1 General

7.1.1 Overview

This test schedule shows the tests and the order in which they shall be carried out as well as the requirements to be met.

Unless otherwise specified, all tests shall be carried out under standard atmospheric conditions for testing as specified in IEC 60068-1, as directed by the applicable part of IEC 60512.

Unless otherwise specified, mated and locked sets of connectors shall be tested. Care shall be taken to keep a particular combination of connectors together during the complete test sequence, i.e. when unmating is necessary for a certain test, the same connector styles as before shall be mated for the subsequent tests.

In the following, a mated and locked set of connector styles shall be called a specimen.

7.1.2 Climatic category

The climatic category and related performance levels are stated in Table 1.

7.1.3 Creepage and clearance distances

The coordination of creepage and clearance distances has to be performed according IEC 60664-1.

See Table 6 for the rated voltage.

See Table 7 for the rated impulse voltage and pollution degree.

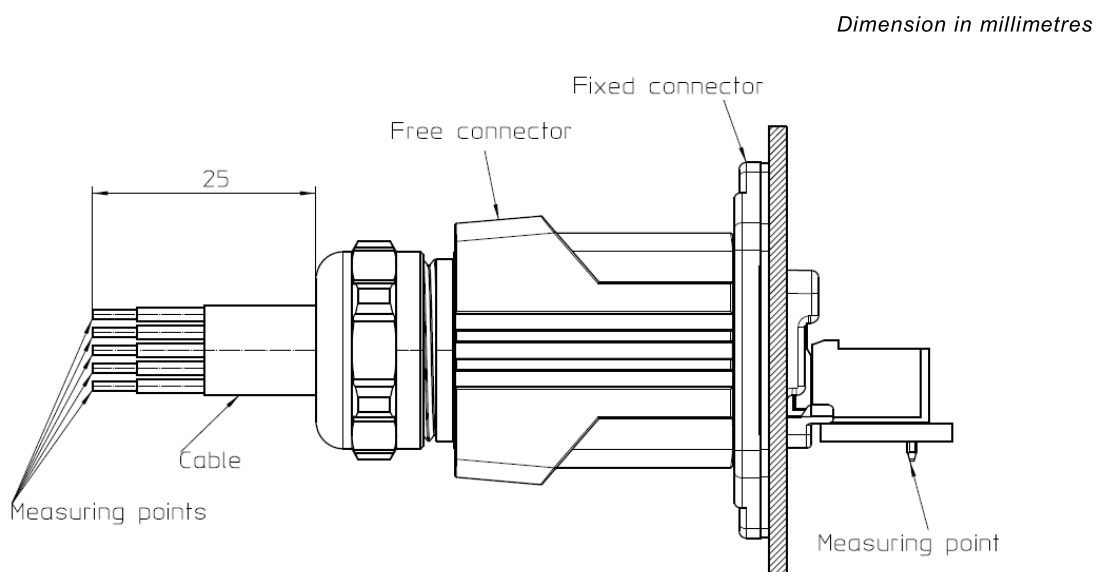
See Table 8 for the voltage proof.

Application information: The permissible rated voltage depends on the application or specified safety requirements. Reductions in creepage or clearance distances may occur due to the printed board or wiring used, and shall duly be taken into account.

7.1.4 Arrangement for contact resistance measurement

Conditions: IEC 60512-2-1, Test 2a
Standard atmospheric conditions
Measuring points, see Figure 8

The measurement of contact resistance shall be carried out on the number of contacts specified. Any subsequent measurements of contact resistance shall be made on the same contacts. Figure 8 shows a contact resistance arrangement.



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Figure 8 – Contact resistance arrangement

7.1.5 Arrangement for dynamic stress tests

Conditions: IEC 60512-6-1, Test 6a, IEC 60512-6-2, Test 6b, IEC 60512-6-3, Test 6c and IEC 60512-6-4, Test 6d.

Figure 7 shows a dynamic stress test arrangement.

7.1.6 Arrangement for testing static load, axial

Conditions: IEC 60512-8-2, Test 8b

7.1.7 Wiring of specimens

The terminations shall be wired, in accordance with the relevant part of IEC 60352. For screw-type or screwless-type connection tests, also the relevant requirements of IEC 60999-1 should be duly taken into account.

7.2 Test schedules

7.2.1 Basic (minimum) test schedule (Table 11)

Table 11 – Basic tests

Test phase	Test			Measurement to be performed		Requirements	
	Title	IEC 60512 Test No.	Severity or condition test	Title	IEC 60512 test No.	PL	All connector styles
1	General examination	1	Unmated connectors	Visual examination Examination of dimensions and mass	1a 1b		There shall be no defect that would impair normal operation The dimensions including contact sequence, creepage and clearance shall comply with those specified
2.1 or 2.2	Engaging and separating forces		Speed: A (mm/s) maximum	Engaging and separating forces or Insertion and withdrawal forces	13a 13b		See 6.5.3
3	Contact resistance		Connection points as in 7.1.4	Contact resistance	2a or 2b		5 mΩ maximum
4	Insulation resistance		Test voltage 500 V ± 15 V d.c. Method A	Insulation resistance	3a		10 ⁸ Ω minimum
5	Voltage proof		Method H	Voltage proof	4a		See Table 8 There shall be no breakdown or flashover
6.1 or 6.2	Soldering Other applicable connection method	12 Series ¹⁾		Contact resistance including termination	2a or 2b		5 mΩ maximum
¹⁾ If the connector features solder terminations, the appropriate test(s) and condition(s) shall be selected from IEC 60512. In case of solderless terminations, other appropriate connection tests shall be additional to, or replace, the specified tests, for example tests of IEC 61076-3, or tests of the applicable parts of IEC 60352, such as test group GP of the full test schedule.							

7.2.2 Full test schedule

7.2.2.1 General

Contact resistance measurements and contact interruption measurements may be performed on different contacts. It is permissible to prepare separate connectors for each test.

Contact resistance measurements and insulation resistance/voltage proof measurements shall be performed on different contacts. It is permissible to prepare separate connectors for each.

Dimensions that affect creepage and clearance distances and other dimensions that affect intermateability shall be measured.

In Table 12 below, the minimum number of contacts shall be satisfied; for example if a connector has only 4 contacts, there shall be 8 connectors for test group AP, to satisfy the minimum number of contacts under test.

Table 12 – Number of test specimens and contacts

Test group	AP	BP	CP	DP	EP	FP	GP	HP	JP	KP
Test specimens	3	3	3	3	3	3	3	3	3	3
Number of contacts	Each specimen fully equipped	Each specimen fully equipped	Each specimen fully equipped	Each specimen fully equipped	Each specimen fully equipped	Each specimen fully equipped	Each specimen fully equipped	Each specimen fully equipped	Each specimen fully equipped	Each specimen fully equipped

For the full test schedule, 30 specimens are needed (10 groups of 3). Within each group, only 2 of the 6 specimens shall be subjected to the IPX5 and IPX7 tests. The same specimens are used for both tests against ingress of water, water jets, IPX5, and temporary immersion, IPX7. Two additional specimens shall be used for the dust-tight test, IP6X.

7.2.2.2 Test group P – Preliminary

The specimens shall consist of the complete connector assembled together.

All specimens shall be subjected to the test group P – preliminary tests in the sequence detailed in the following. Table 13 shows the test group P.

The specimens shall then be divided into the appropriate number of groups. All connectors in each group shall undergo the following tests as described in this detail specification and in the sequence given.

The test parameters required shall not be less than those listed. The following tests specify the characteristics to be checked and the requirements to be fulfilled.

Table 13 – Test group P

Test phase	Test			Measurement to be performed		
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	Requirement All connector styles
P1	General examination	1	Unmated connectors	Visual examination	1a	There shall be no defect that would impair normal operation
				Dimensional examination	1b	The dimensions shall comply with those specified in the relevant figure of Clause 5.
P2	Polarizing method	13e	Not applicable			
P3				Contact resistance	2a	10 mΩ maximum
P4			Test voltage 500 V ± 15 V d.c. Method A8 contacts/specimen	Insulation resistance	3a	Min 10 ⁸ Ω
P5			Contact/contact: Method A mated connectors	Voltage proof	4a	See Table 8 There shall be no breakdown or flashover
			All contacts to test panel: Method A mated connectors			See Table 8 and IEC 60664-1

7.2.2.3 Test group AP – Dynamic/climatic

Table 14 shows the test group AP.

Table 14 –Test group AP

Test phase	Test			Measurement to be performed		Requirements
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	All connector styles
AP1			See 6.5.3	Insertion and withdrawal forces	13b	See 6.5.3
AP2	Gauge retention force		Female contacts only 3 contacts/specimen sizing and retention force gauge See Table 14, 2.1	Engaging and separating forces	16e	Not applicable
AP3	Vibration	6d	10 Hz–500 Hz 0,35 mm or 50 m/s ² 3 axis 10 swaps each axis	Contact disturbance	2e	Duration of disturbance 1 µs max.
				Contact resistance – Millivolt level method	2a	Rise in relation to initial values 5 mΩ max.
				Visual examination	1a	There shall be no defect that would impair normal operation
AP4	Shock	6c	Arrangement according to Figure 7 Half sine shock acceleration 490 m/s ² (50 g) Duration of impact: 11 ms 6 shocks per axis 3 positive axis 3 negative axis	Contact disturbance	2e	Duration of disturbance 1 µs max.
				Contact resistance – Millivolt level method	2a	Rise in relation to initial values 5 mΩ max.
				Visual examination	1a	There shall be no defect that would impair normal operation
AP5	Rapid change of temperature	11d	–25 °C to 70 °C, t = 30 min, 5 cycles	Contact resistance- Millivolt level method	2a	Rise in relation to initial values 5 mΩ max.
				Insulation resistance	3a	10 ⁸ Ω min.
				Voltage proof	4a	See Table 8 There shall be no breakdown or flashover
				Visual examination	1a	There shall be no defect that would impair normal operation

Table 14 (continued)

Test phase	Test			Measurement to be performed		Requirements
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	
AP6	Climatic sequence	11a				
AP6.1	Dry heat	11i	Temp.: 70 °C Duration: 16 h	Insulation resistance at high temperature	3a	10 ⁸ Ω min.
AP6.2	Damp heat, cyclic, first cycle	11m	Method Db Temp.: 40 °C Recovery time: 2 h	Visual examination	1a	There shall be no defect that would impair normal operation
AP6.3	Cold	11j	Temp.: –25 °C Duration: 2 h Recovery time: 2 h	Visual examination	1a	There shall be no defect that would impair normal operation
AP6.4	Damp heat, cyclic, remaining cycles	11m	Conditions according to AP6.2 5 cycles Recovery time: 2 h	Contact resistance – Millivolt level method	2a	Rise in relation to initial values 5 mΩ max.
				Insulation resistance	3a	10 ⁸ Ω min.
				Voltage proof	4a	See Table 8 There shall be no breakdown or flashover
				Insertion and withdrawal forces	13b	See 6.5.3
				Visual examination	1a	There shall be no defect that would impair normal operation
AP7	Impacting water	14g				
AP7.1	IP code second characteristic numeral		See IEC 60529	IEC 60529:1989, Test 14.2.5 and Test 14.2.7		No leakage on contacts
AP7.2	IP code first characteristic numeral		Dust IP6X Test 6 Table 7, see IEC 60529	IEC 60529:1989, Table 7		IP6X no deposit of dust on contacts ^a

Table 14 (continued)

Test phase	Test			Measurement to be performed		Requirements
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	
AP7.3				Contact resistance – Millivolt level	2a	Rise in relation to initial values 5 mΩ max.
					2b	
				Insulation resistance	3a	10 ⁸ Ω min.
				Voltage proof	4a	See Table 8 There shall be no breakdown or flashover
				Insertion and withdrawal forces	13b	See 6.5.3
AP8				Visual examination	1a	There shall be no defect that would impair normal operation
AP9	Polarizing method	13e				It shall be possible to correctly align and mate the appropriate mating connectors. It shall not be possible to mate the connectors in any other than the correct manner
^a It is allowed to perform AP7.2 with an additional specimen, extending the total number of specimen by 1.						

7.2.2.4 Test group BP – Mechanical endurance

Table 15 shows the test group BP.

Table 15 – Test group BP

Test phase	Test			Measurement to be performed		Requirements	
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	PL	All connector styles
BP1				Gauge retention force (optional)	16e		Not applicable
BP2				Engaging and separation forces or Insertion and withdrawal forces (optional)	13a 13b		See 6.5.3
BP3	Locking device mechanical operations (double of the specified number of cycles)		Only for free connectors 2 PL1 or 2 PL2 or 2 PL3 operations of locking device Speed: maximum 20 cycles/min	(optional)		1 2 3	Not applicable
BP4.1	Cable clamp Resistance to cable pull	17c	Only for free connectors with cable glands according to IEC 61984	Visual examination	1a		Table 6 of IEC 61984:2008
BP4.2	Cable clamp resistance to cable torsion	17d		Visual examination	1a		Table 6 of IEC 61984:2008
BP5	Mechanical operation (half of the specified number of cycles)	9a				1 2 3	
BP6				Contact and shield resistance (optional)	2a and 2b		See 6.4.4

7.2.2.5 Test group CP – Moisture

Table 16 shows the test group CP.

Table 16 – Test group CP

Test phase	Test			Measurement to be performed		Requirements	
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	PL	All connector styles
CP1				Engaging and separating forces or Insertion and withdrawal forces	13a 13b		See 6.5.3
CP2	Damp heat, steady state	11c IEC 60068-2-30	21 cycles Low temperature 25 °C High temperature 55 °C Cold humidity 93 % All samples in mated state				
CP3				Insulation resistance			10 ⁸ Ω minimum
CP4				Voltage proof			See Table 8 There shall be no breakdown or flashover
CP5				Contact resistance	2a or 2b		5 mΩ maximum
CP6				Engaging and separating forces or Insertion and withdrawal forces			See 6.5.3
CP7				Visual examination			There shall be no defect that would impair normal operation

7.2.2.6 Test group DP – Heat and electrical load

Table 17 shows the test group DP.

Table 17 –Test group DP

Test phase	Test			Measurement to be performed		Requirements	
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	PL	All connector styles
DP1	Mechanical operations	9a	mm/s max. Rest 5 s (unmated) Fully assembled connectors tested and the locking device of shell is actuated-			1 2 3	100 minimum
DP2	Dry heat or Electrical load and temperature	11 i or 9b	+70 °C Mated connectors 16 h or Duration of test: 500 h, ambient temperature 40 °C, current 3 A, wire cross section 0,5 mm ² , temperature measurement in the middle of specimen, upper temperature limit 70 °C				
DP3				Contact resistance	2a or 2b		5 mΩ maximum
DP4				Insulation resistance	3a		10 ⁸ Ω minimum
DP5				Voltage proof	4a		See Table 8 There shall be no breakdown or flashover
DP6				Visual examination	1a		There shall be no defect that would impair normal operation

7.2.2.7 Test group EP – Dynamic stress

Table 18 shows the test group EP.

Table 18 – Test group EP

Test phase	Test			Measurement to be performed		Requirements	
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	PL	All connector styles
EP1	Vibration-random or Vibration - sinusoidal	6e 6d	Atmospheric conditions, frequency range 10-500 Hz, 0,35 mm, acceleration 5g, duration 2h over 3 axis	Contact disturbance	2e		Duration of disturbance 1 µs max
EP2				Visual examination	1a		There shall be no defect that would impair normal operation
EP3				Contact resistance	2a or 2b		5 mΩ maximum
EP4	Shock	6c	Standard atmospheric conditions, Acceleration 500 m/s ² , Duration 11 ms, 3 shocks in each axis and direction, half-sine wave	Contact disturbance	2e		Duration of disturbance 1 µs max
EP5				Visual examination	1a		There shall be no defect that would impair normal operation
EP6				Contact resistance	2a		5 mΩ maximum

7.2.2.8 Test group FP – Chemical resistivity

Table 19 shows the test group FP.

Table 19 – Test group FP

Test phase	Test			Measurement to be performed		Requirements	
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	PL	All connector styles
FP1.1				Engaging and separating forces	13a		See 6.5.3
FP1.2				or Insertion and withdrawal forces	13b		
FP2	Resistance to fluids	19c	Fluids to be specified between manufacturer and user				
FP3.1				Engaging and separating forces	13a		See 6.5.3
FP3.2				or Insertion and withdrawal forces	13b		
FP4				Contact resistance	2a		5 mΩ maximum
FP5				Insulation resistance	3a		10 ⁸ Ω minimum
FP6				Visual examination	1a		There shall be no defect that would impair normal operation

7.2.2.9 Test group GP – Manufacturing process robustness

Table 20 shows the test group GP.

Table 20 – Test group GP

Test phase	Test			Measurement to be performed		Requirements	
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	PL	All connector styles
GP1.1				Engaging and separating forces	13a		See 6.5.3
GP1.2				or Insertion and withdrawal forces	13b		
GP2.1	Resistance to solder heat ¹⁾ or	12d or 12e		Visual examination	1a		There shall be no defect that would impair normal operation
GP2.2 ¹⁾	to be defined	to be defined					
GP3.1 ¹⁾				Engaging and separating forces	13a		See 6.5.3
GP3.2 ²⁾				or Insertion and withdrawal forces	13b		
GP4 ³⁾				Voltage proof	4a		See Table 8 There shall be no breakdown or flashover
GP5	Contact retention in insert (optional)	15a	3 contacts /specimen				See Table 10
GP6	Cable clamp robustness (optional)	17c or 17d					
GP7				Visual examination	1a		There shall be no defect that would impair normal operation
¹⁾ Other applicable termination tests may be covered by other test sequences. ²⁾ Optional. ³⁾ To be performed only when a resistance to soldering heat test is carried out with contacts installed.							

7.2.2.10 Test group HP – Signal integrity and screening effectiveness tests

Not applicable.

7.2.2.11 Test group JP – Connections

Table 21 shows the test group JP.

Table 21 – Test group JP

Test phase	Test			Measurement to be performed		Requirements	
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	PL	All connector styles
JP1	Applicable tests of connection methods according to IEC 60352 and IEC 60512.						
– JPX	Depending upon the type of solderless termination/connection method, if applicable, a test sequence from the relevant part of IEC 60352 shall be selected.						
NOTE 1 Where test evidence may be presented to the satisfaction of the NSI, confirming that the connection methods used by the connectors have been previously tested in accordance with the specified tests of IEC 60352, and have satisfactorily passed them, test phases JP1 to JPX may be omitted. NOTE 2 If a screw-type clamping unit is described in the standard, its conformance to the requirements to IEC 60999-1, respectively IEC 60999-2, shall be proven.							

7.2.2.12 Test group KP – Additional tests

Table 22 shows the test group KP.

Table 22 – Test group KP

Test Phase	Test			Measurement to be performed		Requirements	
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	PL	All connector styles
KP1	Flexing test	IEC 60998-2-1:2002, Test 10.104	45° from cable entry Time period: 15 min Speed: 10 ± 2 turns per minute Hole diameter: max. 3 mm larger than cable diameter Weight: 4,5 kg	Visual examination	1a		No disabling of the latching mechanism

7.3 Test procedures and measuring methods

The test methods specified and given in the relevant standards are the preferred methods but not necessarily the only ones which can be used. In case of dispute, however, the specified method shall be used.

Where approval procedures are involved and alternative methods are employed it is the responsibility of the manufacturer to satisfy the authority granting approval that any alternative methods which he may use give results equivalent to those obtained by the methods specified.

Unless otherwise specified, all tests shall be carried out under standard atmospheric conditions for testing as specified in IEC 60068-1.

7.4 Pre-conditioning

Before the tests are performed, the connectors shall be preconditioned under conditions specified in IEC 60068-1 for a period of 24 h, unless otherwise specified by the detail product specification or by the manufacturer.

7.5 Wiring and mounting of specimens

7.5.1 Wiring

Where wiring of test specimens is required, the detail product specification shall contain sufficient information to perform the tests.

7.5.2 Mounting

When mounting is required in a test, the connectors shall be rigidly mounted on a metal plate, a printed board or to specified accessories, whichever is applicable, using the normal mounting method, fixing devices and panel cut-out as laid down in the detail product specification.

When mounting is required in a test, unless otherwise specified, the connectors shall be rigidly mounted on a metal plate or to specified accessories, whichever is applicable, using the specified connection methods, fixing devices and panel cut-outs as laid down in 5.7.

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